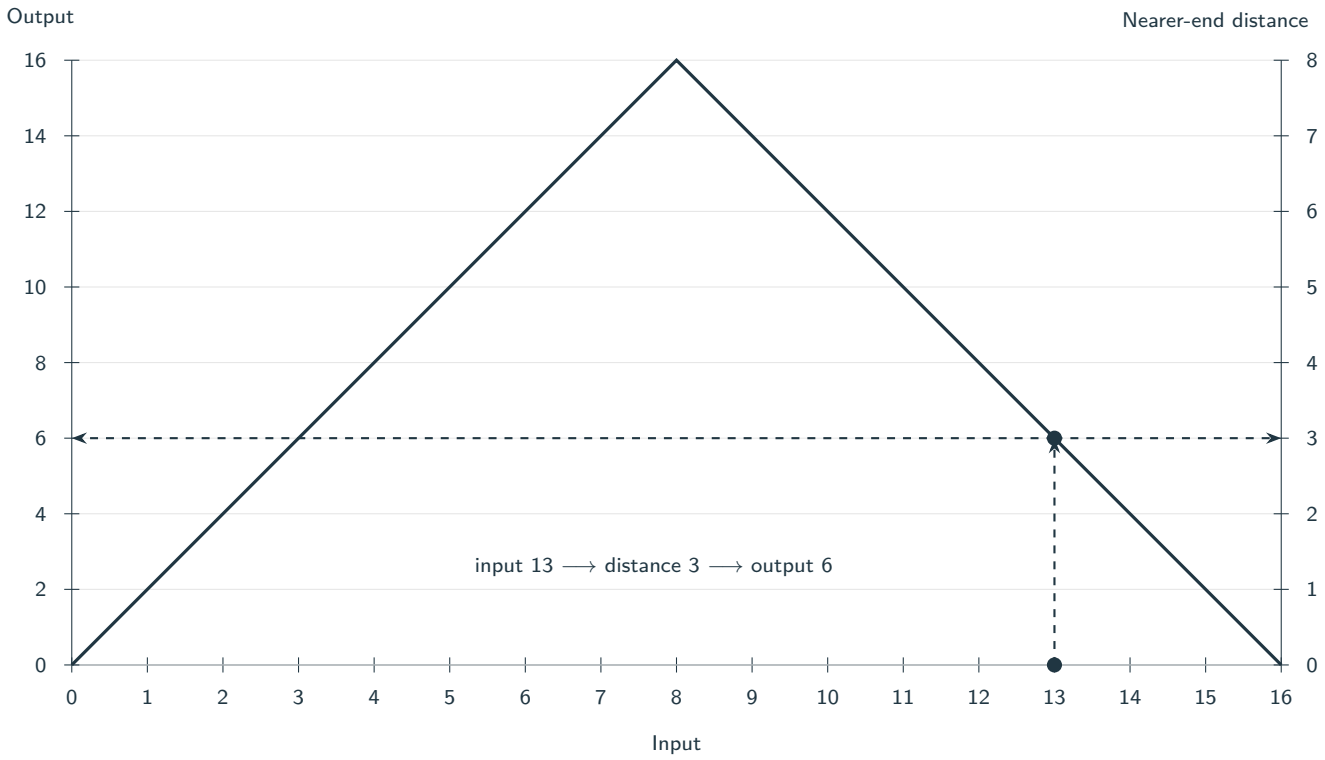
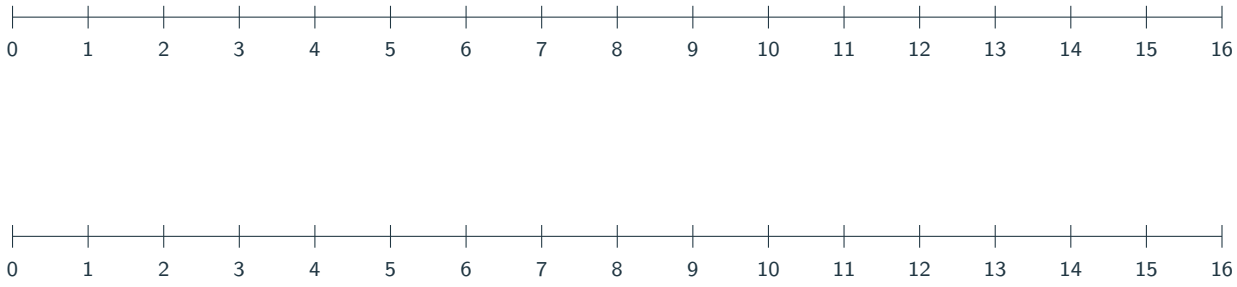
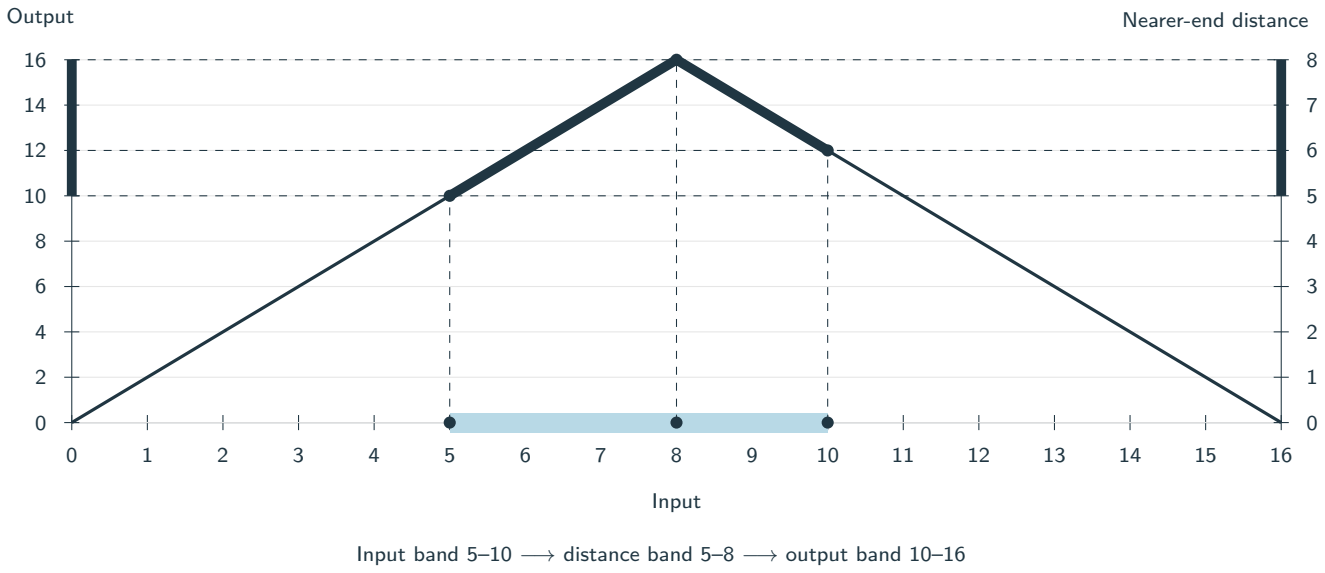
One move: find the nearer-end distance in track units, then double it. On the map, go up from your input to the sloping line, then across to read the distance and output. Put the marker at the output on the 0–16 track. Points between ticks are allowed.



Problem 1: Put two markers at different whole-number starts. Find every pair that meets after one move. Then find two starts whose markers move farther apart for three moves in a row.

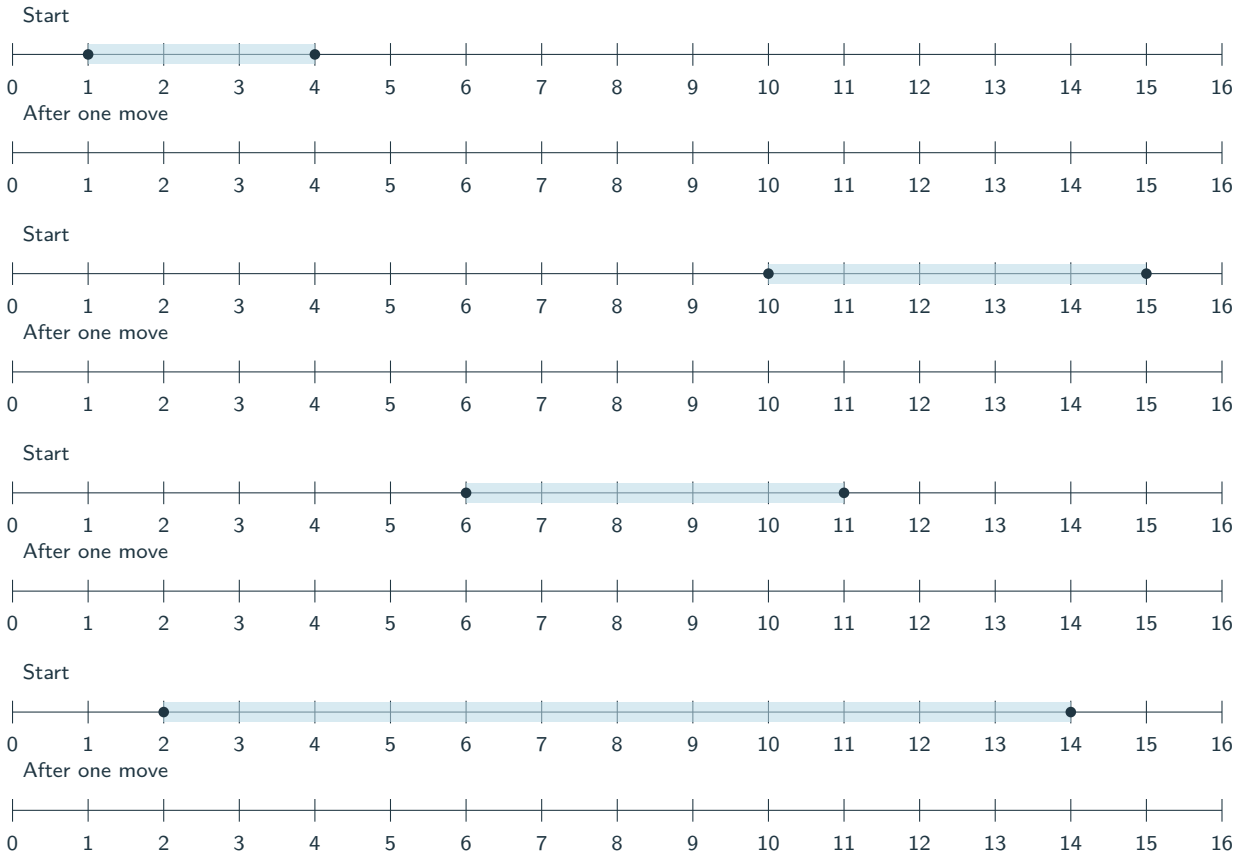


A band includes every position between its ends, including the ends. Move all of them through the map. One marker follows the rule from its unknown start; it is never placed at a fresh start.



Input band 5–10 ⟶ distance band 5–8 ⟶ output band 10–16

Problem 2: Shade all possible places after one move for each starting band.



Problem 3: Start with each band below. Move the whole band until every place on the track is possible. How many moves does each need?

